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ORIGINAL ARTICLE

TIM-3 blockade combined with bispecific antibody MT110 enhances the anti-tumor effect of γδ T cells

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Abstract
As ideal cells that can be used for adoptive cell therapy, γδ T cells are a group of homogeneous cells with high proliferative and tumor killing ability. However, γδ T cells are apt to apoptosis and show decreased cytotoxicity under persistent stimulation in vitro and cannot aggregate at tumor sites efficiently in vivo, both of which are two main obstacles to tumor adoptive immunotherapy. In this study, we found that the immune checkpoint T-cell immunoglobulin domain and mucin domain 3 (TIM-3) were up-regulated significantly on γδ T cells during their ex vivo expansion and this up-regulation contributed to the dysfunction of γδ T cells. Although the killing ability of γδ T cells against breast cancer cells which exhibited a high level of epithelial cell adhesion molecule (EpCAM) was enhanced, the level of TIM-3 on γδ T cells was also further up-regulated under the application of the bispecific antibody MT110 (anti-CD3 × anti-EpCAM) which can redirect T cells to target cells. Besides, these γδ T cells with up-regulated TIM-3 exhibited an increased susceptibility to apoptosis. By reinvigorating dysfunctional γδ T cells and promoting them to accumulate at tumor sites, the combined use of TIM-3 inhibitor and MT110 could further enhance the anti-tumor effect of the adoptively transfused γδ T cells. These results may have clinical implications for the design of new translational anti-tumor regimens aimed at combining checkpoint blockade and immune cell redirection.

Keywords γδ T cells · Cytotoxicity · Apoptosis · Epithelial cell adhesion molecule · Bispecific antibody · T-cell immunoglobulin domain and mucin domain 3

Introduction

Nowadays, the adoptive cell therapy (ACT) has been attempted as an alternative treatment option for tumor patients to eliminate residual tumor cells, boost tumor immunity, and increase drug sensitivity. ACT is defined as the transfusion of autologous or allogeneic T cells, NK cells, dendritic cells (DC), or any other immune cells into tumor-bearing hosts or cancer patients, considering that these cells can help to control tumor growth [1]. It has made rapid progress and immunotherapy has been recognized as the fourth anti-tumor modality after surgery, chemotherapy, and radiotherapy [2]. Utilizing non-specific immune cells such as CIK (cytokine induced killer cells), NK and γδ T cells plays an important role in ACT, since these immune cells can efficiently kill tumor cells and are controllable and safety in vivo. Compared to the heterogeneous CIK cells, γδ T cells are a group of relatively pure T cells whose TCRs are composed of γ and δ chains. Unlike αβ T cells which recognize peptidic antigens in the context of highly polymorphic MHC molecules, γδ T cells recognize low-molecular-weight phosphate containing highly conserved non-peptidic molecules presented in an MHC-independent manner [3]. Vγ9Vδ2 subset, the major γδ T-cell subset existed in peripheral blood, can recognize phosphoantigens such as isopentenyl pyrophosphate (IPP) produced in mammalian cells through the mevalonate pathway. Therefore, as a synthetic analog of

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IPP, bromohydrin pyrophosphate (BrHPP) has been widely used to amplify Vγ9Vδ2 T cells in vitro [4]. Moreover, these cells can also be activated by aminobisphosphonates such as zoledronate, since these materials can inhibit the farnesyl pyrophosphate synthase, an enzyme of the mevalonate pathway, leading to an accumulation of IPP [5, 6]. Comparative BrHPP and zoledronate experiments with donor peripheral blood mononuclear cells (PBMC) showed similar Vγ9Vδ2 T-cell rate at the end of the culture [7]. γδ T cells used in this study are Vγ9Vδ2 cells that induced from peripheral blood using zoledronate in combination with IL-2.

So far, although immune therapy by transfusing T cells or NK cells has made certain achievement, there are still two pivotal problems limited the tumor therapy efficacy. One of the problems is that T cells are easily becoming dysfunctional under excessive stimulation and the other is that the adoptive transfused immune cells are difficult to infiltrate and accumulate in tumor sites in vivo.

Understanding the basic mechanisms of dysfunction of ex vivo expanded γδ T cells is crucial for clinical applications. These dysfunctional statuses are similar to T-cell exhaustion which is characterized by the stepwise and progressive loss of T-cell function such as proliferation ability, cytotoxicity, and cytokine secretion in response to antigen stimulation and can culminate in the physical deletion of the responding cells [8]. T-cell exhaustion is often associated with the inhibitory signals mediated by immune checkpoints which consist of inhibitory and stimulatory pathways that maintain self-tolerance and assist with immune response [8]. The identification of the importance of inhibitory receptors in the dysregulation of cellular immune responses in tumor immunity has revealed new potential therapeutic targets for restoring anti-tumor immune reaction [9]. The most broadly studied checkpoints are the inhibitory pathways consisting of cytotoxic T lymphocyte-associated molecule 4 (CTLA-4), programmed cell death receptor-1 (PD-1), and programmed cell death ligand-1 (PD-L1). New immune checkpoint such as lymphocyte activation gene-3 (LAG-3), B- and T-cell lymphocyte attenuator (BTLA, CD272), T-cell immunoglobulin domain and mucin domain 3 (TIM-3), T-cell immunoreceptor with Ig and ITIM domains (TIGIT), and their mediated inhibitory pathways are under investigation [10, 11]. By examining the expression level of some immune checkpoints on ex vivo expanded γδ T cells, we found that TIM-3 was up-regulated significantly during γδ T-cell induction and expansion process. TIM-3 was identified as an immune checkpoint in 2002 and mainly expressed on interferon (IFN)-γ-producing $CD4^+$ T helper type 1 (Th1) and on $CD8^+$ T cytotoxic type 1 (Tc1) cells [12–14]. The structure of TIM-3 is somewhat reminiscent of that of mucosal address in cell adhesion molecule-1, which contains two immunoglobulin domains and a mucin-rich region [12]. The S-type lectin galectin-9 (Gal-9) was then identified as a TIM-3 ligand and Gal-9 triggering of TIM-3 on Th1 cells has been shown to induce cell death [15]. Thus, TIM-3 came to be known as an important negative regulatory molecule for abrogating Th1- and Tc1-driven immune responses.

Although immune cells such as CIK, NK, and γδ T cells could efficiently kill tumor cells in many ex vivo experiments, adoptively transferring of these cells to tumor patients has a limited clinical outcome. An important reason is that the adoptively transfused immune cells could not efficiently migrate to tumor sites and infiltrate in it. With the help of bispecific antibodies, this problem is expected to be better solved. By binding to immune cells and tumors simultaneously, bispecific antibodies can efficiently redirect immune cells to tumor cells. Due to its effective tumor therapeutic effects from a series of clinical trials, the bispecific antibody, catumaxomab, whose binding targets are CD3 and EpCAM, has been approved by the European Medicine Agency (EMA) for the treatment of malignant ascites in patients with EpCAM-positive carcinomas, where standard therapy is not available or no longer feasible [16, 17]. Recently, advances in bioengineering and expanded insight in tumor immunology have resulted in the emergence of novel bispecific antibody (bsAb) constructs that are capable of recruiting immune effector cells to the tumor microenvironment. As one such type of bsAbs, bispecific T-cell engager (BiTE®) molecules, which are made by fusing an anti-CD3 to an anti-tumor-associated antigen (TAA) single-chain variable fragment (scFv) via a 5 residual peptide linker, has been extensively studied. They can effectively recruit and activate polyclonal T cells, resulting in a highly cytotoxic response in the absence of co-stimulation signals [18]. The bispecific antibody MT110 (anti-CD3 × anti-EpCAM) which we used in this study is a BiTE® that lacks of Fc region when compared with catumaxomab. With reduced structure and weight, it could penetrate to tumor site more easily and promote γδ T cells to accumulate in tumor sites in vivo.

In this study, we found that the up-regulated immune checkpoint TIM-3 during γδ T-cell induction and expansion in vitro was correlated with γδ T-cell dysfunction. Blocking of this checkpoint by TIM-3 inhibitor could efficiently prevent γδ T-cell apoptosis and reinvigorate the dysfunctional γδ T cells. Besides, with the help of the bispecific antibody MT110, the cytotoxicity of γδ T cells with TIM-3 blockade was further strengthened and these cells could accumulate in tumor sites more efficiently. Furthermore, adoptively transfuse γδ T cells in combination with these two antibodies to breast cancer mice exhibited a better anti-tumor effect. This studied target and immunotherapeutic strategy will also shed new light on the treatment of other types of tumors.

Materials and methods

γδ T-cell induction and expansion

Blood and leukopheresis were taken from healthy volunteers ($n = 5$) and breast cancer patients ($n = 5$), and written informed consent was obtained from all the patients and healthy subjects prior to blood collection. Clinical information including age, gender, and tumor stage is collected and listed in Table 1. This study was approved by the research ethics committee of Qingdao University and confirmed to the provisions of the Declaration of Helsinki. γδ T cells were generated from PBMC with Ficoll (GE Healthcare, Uppsala, Sweden) separation. Briefly, PBMC were stimulated with 1 μM zoledronate (Novartis, Basel, Switzerland) and cultured overnight in RPMI 1640 at a cell density of 1.5×10^6/ml supplemented with 10% heat-inactivated autologous plasma and recombinant human IL-2 (400 U/ml; Quangang, Jinan, China). After culturing for about 4 or 5 days, fresh medium with IL-2 (200 U/ml) was added to the culture system every 2 or 3 days according to the specific conditions.

Mice, tumor cell lines, and tumor inoculation

Female nude BALB/c mice (4 weeks of age) were purchased from Shanghai Slac Animal Inc. (Shanghai, China). All mice were bred in a specific pathogen-free barrier facility in Experimental Animal Center of Qingdao University (Qingdao, China). Human breast cancer cell line MDA-MB-231 maintained in L15 supplemented with 10% heat-inactivated fetal calf serum (FCS), while human chronic myelogenous leukemia cell line K562 maintained in complete RPMI1640 supplemented with 10% heat-inactivated FCS and incubated at 37 °C in humidified air with 5% CO_2. Human neuroblastoma cell line SH-SY5Y was maintained in DMEM supplemented with 10% heat-inactivated fetal calf serum (FCS) and incubated at 37 °C in humidified air with 5% CO_2. Cells were cultured to cover 70–80% of the flasks. Trypsin–EDTA (Corning, USA) was used for digestion, counting, and passage. MDA-MB-231 cells were harvested for inoculation of mice with 0.05% trypsin–EDTA, washed twice, and resuspended in PBS. BALB/c nude mice were inoculated subcutaneously (s.c.) in the flank with 5×10^6 MDA-MB-231 cells.

Flow cytometry analysis and isolation of γδ T cells

PBMC were cultured in the γδ T-cell culture system and collected at different points in culturing time. After washing twice with phosphate-buffered saline (PBS), the cells were double or triple stained with fluorescein isothio-cyanate (FITC)-anti-human CD3, phycoerythrin (PE)-anti-human TCR Vδ2, peridinin–chlorophyll–protein complex (PerCP)-anti-CTLA4, PerCP-anti-PD1, and PerCP-anti-TIM3; FITC-anti-human CD25, FITC-anti-human HLA-DR, and PerCP-anti-human CD69 (BD Biosciences, USA). Isotype-matched control antibodies were used for all analyses. The cells (1×10^6) were incubated with various conjugated mAbs for 15 min at room temperature. For intracellular detection of IFN-γ and perforin, γδ T cells were fixed/permeabilized (Biolegend, USA) and washed twice with stain buffer. Then, the cells were stained with APC-anti-human IFN-γ or perforin (Biolegend, USA) according to the protocol. After incubation, cells were washed twice and then analyzed using

Table 1 Ex vivo expansion of γδ T cells from healthy donors and breast cancer patients

Case number	Age/gender	Tumor stage	Day 0		Day 21			Cytotoxicity (%)
			PMBC seeded ($\times 10^7$)	γδ T cells (%)	Cells recovered ($\times 10^7$)	γδ T cells (%)	Expansion fold	
HD1	45/F	–	1	1.9	21	81.8	904	18.8
HD2	49/F	–	1	3.3	18	91.6	500	13.6
HD3	55/F	–	1	3.6	27	90.3	677	16.1
HD4	43/F	–	1	2.0	25	84.6	1057	14.0
HD5	53/F	–	1	2.6	25	88.3	849	16.9
BCP1	61/F	II_A	1	4.0	19	86.4	410	13.9
BCP2	51/F	III_A	1	2.8	27	77.0	743	12.6
BCP3	55/F	II_B	1	2.6	27	89.7	932	17.5
BCP4	48/F	II_A	1	2.5	24	91.5	878	17.0
BCP5	49/F	II_A	1	3.1	20	82.2	530	15.3

The expansion fold = number of γδ T cells recovered/number of γδ T cells seeded. γδ T cells of 15 days' expansion from different individuals were co-cultured with MDA-MB-231 cells at an effector:target ratio of 1:1 for 12 h and their cytotoxicity was determined by flow cytometry

HD healthy donor, *BCP* breast cancer patient, *F* female, *M* male

FACS Calibur (BD Biosciences, USA) and data on 10,000 cells were acquired and analyzed by Cellquest software. The lymphocytes were gated for forward/side scatter.

For isolation of Vδ2$^+$ T cells, cells in the γδ T-cell culture system were collected and labeled with anti-human TCR Vδ2-biotin and anti-biotin MicroBeads (Miltenyi, Germany). Subsequent positive selection was carried out using MS columns (Miltenyi) according to the manufacturer's protocol. Isolated Vδ2$^+$ T cells were cultured in RPMI1640 medium supplemented with 10% FCS and 200 IU/ml IL-2.

Apoptosis assay

The isolated γδ T cells were cultured in RPMI 1640 at a cell density of 1.5×10^6/ml supplemented with 10% heat-inactivated autologous plasma. Galectin-9 or agonistic TIM-3 antibody (Clone: 344801, Catalog: MAB23651, R&D systems, USA) was added to each group to obtain a final concentration of 40 pg/ml or 1 μg/ml, respectively. After being treated by galectin-9 or agonistic TIM-3 antibody for about 24 h, cells were washed twice and then apoptosis was quantified by annexin V-FITC and propidium iodide (PI)-double staining according to a staining kit (Sigma, USA). Briefly, cells were washed twice with PBS and then 400 μl binding buffer containing 5 μl annexin V-FITC was added. Following gentle vortex, the mixture was incubated for 15 min at 4 °C in the dark and then 10 μl PI was added. Following gentle vortex, the samples were analyzed using an FACS Calibur flow cytometer within 15 min.

Blocking studies

To investigate the effect of TIM-3 blockade on γδ T cells, γδ T cells were first pre-incubated for 20 min with TIM-3 inhibitor (anti-human-TIM-3 blocking antibody, α-TIM-3) (Clone: F39-2E2, Catalog: 345012, Biolegend, USA) or isotype-matched antibody with the final concentration of 5 μg/ml in the corresponding groups. For the TIM-3-Fc group, TIM-3-Fc (Biolegend, USA) was added to obtain a final concentration of 2 μg/ml. Then, galectin-9 (Biolegend, USA) was added to all the groups to obtain a final concentration of 40 pg/ml. After culturing for 24 h, apoptosis of γδ T cells in each group was examined using the method mentioned above. To explore the function of TNF-α on the regulation of TIM-3 on γδ T cells, γδ T cells (or cells in the γδ T and MDA-MB-231 co-culture system) were treated by TNF-α inhibitor (anti-TNFRI blocking antibody, a final concentration of 0.3 μg/ml) (R&D systems, USA) or NF-κB inhibitor (MG132, a final concentration of 3 μM) (R&D systems, USA) in the presence of recombinant human TNF-α protein (a final concentration of 3 ng/ml) (R&D systems, USA). After culturing for 24 h, the expression level of TIM-3 on γδ T cells in each group was examined by an FACS Calibur flow cytometer.

Cell proliferation assay

Cell proliferation was assessed by staining with carboxyfluorescein diacetate succinimidyl ester (CFSE, Invitrogen, USA). Isolated γδ T cells were stained with CFSE prior to galectin-9 (40 pg/ml) or agonistic TIM-3 antibody (1 μg/ml) treatment. Then, the labeled cells were analyzed by flow cytometry after 4 day culture. Briefly, after 8–10 days of induction and proliferation, γδ T cells in the γδ T-cell culture system were collected, washed, and resuspended in 1 ml PBS with 1% BSA at a final concentration of 5×10^6 cells/ml, and then labeled for 10 min at 37 °C with CFSE (2 μM). The staining was stopped by incubation on ice for 5 min in five volumes of ice-cold complete medium. The cells were then washed three times with ice-cold PBS containing 1% BSA and returned to culture under appropriate conditions. CFSE staining was analyzed using an FACS Calibur.

Western blotting

To confirm the accuracy of apoptosis and proliferation of γδ T cells upon TIM-3 activation, the Western blotting was administrated in expanded γδ T cells. The γδ T cells treated by various materials for 24 h were lysed with cold lysis buffer and centrifuged at 12,000*g* for 10 min; loading buffer were mixed with cell lysates and boiled for 5 min at 95 °C. 10 μg total protein was loaded into 10% discontinuous SDS-polyacrylamide gel for electrophoresis and then transferred onto polyvinylidene fluoride membranes. Membranes were blocked in 5% bovine serum albumin and incubated with primary antibodies (anti-caspase 3, anti-cleaved caspase 3, anti-cyclin B1, anti-cyclin D1, and anti-β-tubulin) (Abcam) overnight at 4 °C in Tris buffered saline with 0.1% Tween20. After washing, immunostaining was performed using appropriate secondary antibody at a dilution of 1:10,000 and developed with ECL plus Western blot detection system (Azure Biosystems, USA).

Co-culture of γδ T cells and tumor cells

γδ T cells were co-cultured with K562, SH-SY5Y or MDA-MB-231 cells at effector:target ratios of 10:1, 5:1, and 1:1 in 1 ml of medium with IL-2 (100 U/ml). γδ T cells in some of the co-cultural groups were first pre-incubated for 20 min with TIM-3 inhibitor or isotype-matched antibody at 5 μg/ml. For other co-cultural groups, TIM-3-Fc was added to obtain a final concentration of 2 μg/ml. There were also three control groups containing only target cells (K562, SH-SY5Y or MDA-MB-231 cells). For the α-CD3 (OKT3) (ebioscience, USA) or MT110 treatment experiment, γδ T cells were co-cultured with SH-SY5Y or MDA-MB-231 cells at an effector:target ratio of 5:1. Then, a series dose of MT110 was added to the co-culture system to obtain a final concentration of 0, 0.1, 1, 10, 30, 50, 100, and 1000 ng/ml or α-CD3 was added to the co-culture system to obtain a final concentration of 30 ng/ml. After 12 h of co-culture, we collected the supernatant of each culture system for later cytokine determination.

Cytotoxic assay

γδ T cells were used as effector cells, while tumor cells were used as target cells. Tumor cells were labeled with CFSE and then co-cultured with γδ T cells as mentioned above. The control target cells with only CFSE labeled were cultured in the same culture system. Plates were gently mixed and incubated at 37 °C in 5% CO_2 for 12 h. At the end of the incubation period, we collected all cells in each well into tubes, respectively, and all tubes were placed on ice. 20 µl of PI (1 µg/ml) were added to each tube for 10 min on ice. Finally, 100 µl of complete medium was added before acquisition on an FACS Calibur. The calculation of cytotoxicity was based on the degree of reduction of viable target cells with the ability to retain CFSE and exclude PI ($CFSE^{high}PI^{-}$).

Analysis of γδ T-cell infiltration and accumulation in vivo

After 10 days of MDA-MB-231 inoculation, we selected 18 mice with similar tumor volumes and were randomly divided into 3 groups: γδ T ($n=6$), γδ T and MT110 ($n=6$) and γδ T and α-TIM-3 and MT110 ($n=6$) groups. Before adoptive transfusion of γδ T cells, we stained these cells with PKH26 (Sigma, USA) according to the manufacture. Then, we dyed some of these cells with Trypan blue (Solarbio, China) to examine their viability. 5×10^6 γδ T cells conjugating with or without α-TIM-3 (15 µg) and/or MT110 (2 µg) in 0.3 ml PBS supplemented with rhIL-2 (200 U) were intravenously injected into the tail of each mouse in the corresponding group, respectively. After 48 h, we injected α-TIM-3 and/or MT110 to each mouse of the corresponding group. 24 or 120 h after γδ T-cell transfusion, the blood (50 µl) from the orbital veins of these mice was collected into an anti-clotting solution (5 mM EDTA in PBS), washed with PBS, resuspended in 100 ml PBS, and detected the proportion of PKH26-positive cells by cytometry. The remaining blood was collected in tubes and centrifuged for 10 min at 1000×*g* within 30 min of collection. The separated supernatant and plasma were stored at − 80 °C for later cytokine determination. Then, the tumor tissue was immediately removed from these tumor mice, embedded in optimal cutting temperature medium, and snap frozen. Serial horizontal cryosections 10 µm in thickness were placed on silane prep slides. Then, the slides were air dried and fixed in 95% ethanol at room temperature for 20 min. 95% ethanol-fixed sections were air dried and stained with DAPI (Solarbio, China) according to the manufacture. To quantify PKH26-positive cells, slides were scanned and analyzed with the Panoramic Viewer software (3D Histech, Hungary). For each quantification, cells were counted in six microscopic fields (software magnification 40×) and the average was used for analysis. To allow for an irregular distribution of the cells, three microscopic fields were randomly chosen adjacent to the tumor capsule and in the central part of the tumor each.

Cytokine determination

Twenty heathy volunteers and twenty-eight cancer patients who had been diagnosed with breast cancer at the Second Affiliated Hospital of Qingdao University were enrolled. Clinical information including age, gender, and tumor stage at presentation is collected and listed in Table 2. This study was approved by the research ethics committee of Qingdao University and confirmed to the provisions of the Declaration of Helsinki. The supernatant of the MDA-MB-231 cell culture system and MDA-MB-231 and γδ T-cell co-culture system, and the plasma of the enrolled volunteers and breast cancer patients was collected and centrifuged for 10 min at 800×*g*. After centrifugation of the supernatant and plasma collection mentioned above, galectin-9 in the supernatant of culture system and human serum, TNF-α and IFN-γ in co-culture system and serum of tumor-bearing mice were determined by ELISA using the human galectin-9, TNF-α and IFN-γ ELISA Kit (Cloud–Clone Corp, Houston, USA) according to the manufacturer's instructions.

Table 2 Cases' characteristics

Characteristics	No. of HD	No. of BCP
Age		
< 41	4	4
41–60	14	18
> 60	2	6
Gender		
Male	0	0
Female	20	28
Tumor stage		
II_A	–	13
II_B	–	6
III_A	–	4
IV	–	5

HD healthy donors, *BCP* breast cancer patients

In vivo xenograft experiments

After 10 days of MDA-MB-231 inoculation, we selected 25 mice with similar tumor volumes and were randomly divided into five groups: PBS control ($n=5$), γδ T ($n=5$), γδ T and α-TIM-3 ($n=5$), γδ T and MT110 ($n=5$), and γδ T and α-TIM-3 and MT110 ($n=5$) groups. On day 12, day 15, and day 18 after tumor cell inoculation, 5×10^6 γδ T cells conjugating with or without α-TIM-3 and/or MT110 in 0.3 ml PBS supplemented with rhIL-2 (200 U) were intravenously injected into the tail of the mice in the appropriate group, respectively. The mice were monitored daily and the appearance of a hard texture at the inoculation site was identified as a tumor nodule. The tumor volumes were measured every other day using digital calipers and were calculated using the formula $(A\times B^2)\times0.5$, in which A was the largest and B was the shortest dimension. When the largest diameter of the tumor reached 2 cm or the mice experienced severe ulcers, the mice were sacrificed.

Statistical analysis

Data are shown as mean ± standard deviation (SD) for one representative experiment. Statistical significance was determined by Student's t test or ANOVA, where suitable and a value of $P<0.05$ ($*p<0.05$; $**p<0.01$) was considered to be statistically significant. All the experiments described were performed at least three times to warrant the reproducibility of the results.

Results

γδ T cells became dysfunctional at late stage of ex vivo expansion

Using zoledronate and IL-2, we successfully induced γδ T cells from PBMC isolated from five breast cancer patients and five healthy volunteers. All the γδ T cells originated from different individuals had a robust expansion ex vivo and the proportion of expanded $CD3^+V\delta2^+$ T cells increased to about 85% after 20 day culture. γδ T cells from both healthy donors and breast cancer patients exhibited a similar expansion capacity and tumor cell killing ability (Table 1). A representative analysis of the proportion of $CD3^+V\delta2^+$ T cells expanded from a breast cancer patient is presented in Fig. 1a. At the same time, we examined the activation status of these cells by detecting the dynamic expression of the activation markers such as CD25, CD69, and HLA-DR on $V\delta2^+$ T cells. Results showed that the expression of CD25, CD69, and HLA-DR increased progressively and reached a maximum value after 20 day culture, indicating that γδ T cells had been fully activated (Fig. 1b, c). We also examined the dynamic cytotoxicity of the isolated γδ T cells ($V\delta2^+$ T cells) by co-culturing with different cancer cell lines such as K562, SH-SY5Y, and MDA-MB-231 at different ratios. We found that the highest γδ T cells killing efficacy on these cell lines occurred at day 15, and then, the cytotoxicity of γδ T cells began to decrease gradually along with the culturing time lasting (Fig. 1d). We speculated that this impaired cytotoxicity after day 15 might be due to γδ T-cell dysfunction which is similar to T-cell exhaustion characterizing by apoptosis and/or impaired effector function under chronic infection or cancer [8]. Therefore, we examined the viability of the isolated $V\delta2^+$ T cells (expanded from two healthy donors and three breast cancer patients) in the γδ T-cell culture system at different time points such as day 10, day 14, day 18, and day 22. As expected, the proportion of Annexin V positive $V\delta2^+$ T cells increased stably with the prolongation of culturing time (Fig. 1e). These data indicated that although γδ T cells could be efficiently expanded and fully activated ex vivo, they became dysfunctional gradually at the late stage.

Immune checkpoint TIM-3 had a robust up-regulation during ex vivo γδ T-cell expansion

It is known that many factors contribute to immune cell exhaustion; however, immune checkpoint regulators are the most important one, especially for the activated T cells. Immune checkpoints refer to a series of inhibitory molecules expressed on immune cells and activation of these checkpoints can limit excessive immune responses. Therefore, these immune checkpoints play an important role in keeping immune balance. To investigate the correlation between γδ T-cell dysfunction and immune checkpoints, we examined the dynamic expression of the recent extensively studied immune checkpoints such as CTLA-4, PD-1, and TIM-3 on γδ T cells expanded from three healthy donors and three breast cancer patients. Phenotypic analysis could not reveal any significant differences between the expanded γδ T cells from healthy donors and breast cancer patients (data not shown). A representative analysis of the dynamic expression of these three immune checkpoints on γδ T cells expanded from a breast cancer patient is presented in Fig. 1f. The results showed that although CTLA-4 were rapidly up-regulated and reached the maximum value on day 4, its expression gradually decreased to a very low level in the following days and its expression level was almost not detected at the late stage. In addition, the expression of PD-1 was always kept a low level during the whole period of γδ T-cell induction and expansion in vitro, although it experienced a slight up-regulation from day 9 to day 18. Of note, TIM-3 had a rapid up-regulation at the early stage of the induction and kept a robust high level during the long-term expansion of γδ T cells. These results suggested that TIM-3 might act as

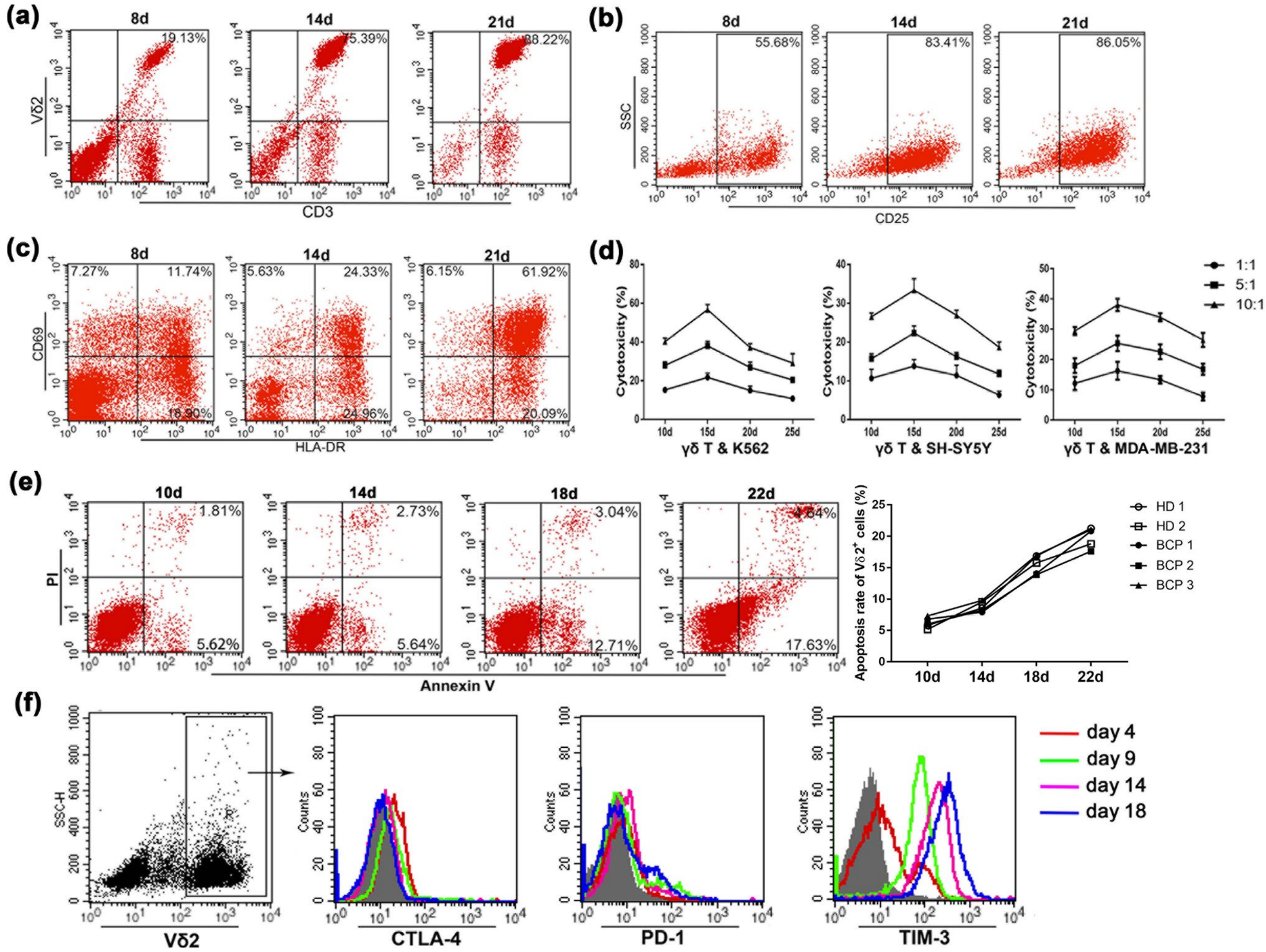

Fig. 1 γδ T cells became dysfunctional and had a robust up-regulation of TIM-3 during ex vivo induction and expansion. Cells in the γδ T-cell culture system were collected at day 8, day 14, and day 21 and the proportion of $CD3^{+}V\delta 2^{+}$T cells **a** and their activation markers such as CD25 **b**, CD69 and HLA-DR **c** at the corresponding time were determined by flow cytometry. **d** $V\delta 2^{+}$T cells (isolated at day 10, day 15, day 20 and day 25 from the γδ T-cell culture system) were co-cultured with K562, SH-SY5Y and MDA-MB-231 at different ratios (1:1, 5:1 and 10:1), respectively, for 12 h, and then, their cytotoxicity was determined by flow cytometry. **e** Representative annexin V/PI staining demonstrates increasing frequencies of the apoptosis of the isolated $V\delta 2^{+}$T cells in the γδ T-cell culture system at day 10, day 14, day 18, and day 22. The apoptosis rate of γδ T cells expanded from 2 healthy donors and 3 breast cancer patients was calculated as the numbers of the first plus fourth quadrants divided by those of all the four quadrants of the dot plot diagrams. Each symbol represents one individual donor (HD: Healthy donor; BCP: breast cancer patient). **f** The expression level of CTLA-4, PD-1, and TIM-3 on $V\delta 2^{+}$T cells in the γδ T-cell culture system at day 4 (red line), day 9 (green line), day 14 (pink line), and day 18 (blue line) was detected by flow cytometry. The filled histograms represent isotype controls staining and the open ones are specific staining. Similar results were obtained in three independent experiments and shown is representative data

an important inhibitory molecule for γδ T cells, since TIM-3 had a rapid up-regulation and thereafter kept a very higher level in the process of γδ T-cell induction and expansion, while the expression of CTLA-4 or PD-1 was not changed markedly.

Interaction of TIM-3 and TIM-3 ligands induced apoptosis of γδ T cells

TIM-3 was first identified as a molecule that is specifically expressed by terminally differentiated Th1 cells, and it was shown to lead to the death of effector Th1 cells on binding to its ligand, galectin-9 (also known as LGALS9) [12, 15]. Galectin-9, a β-galactoside-binding lectin, is widely distributed in different tissues and regulates cell differentiation, adhesion, aggregation, and cell death [19]. Galectin-9 can be found both intra- and extracellularly. However, the secretory pathway of galectin-9 has not been fully understood so far [20]. In this study, we found that galectin-9 was preferentially localized in the cytoplasm of γδ T cells, SH-SY5Y cells, and MDA-MB-231 cells (Fig. 2a). However, the expression of galectin-9 was nearly not detected on the

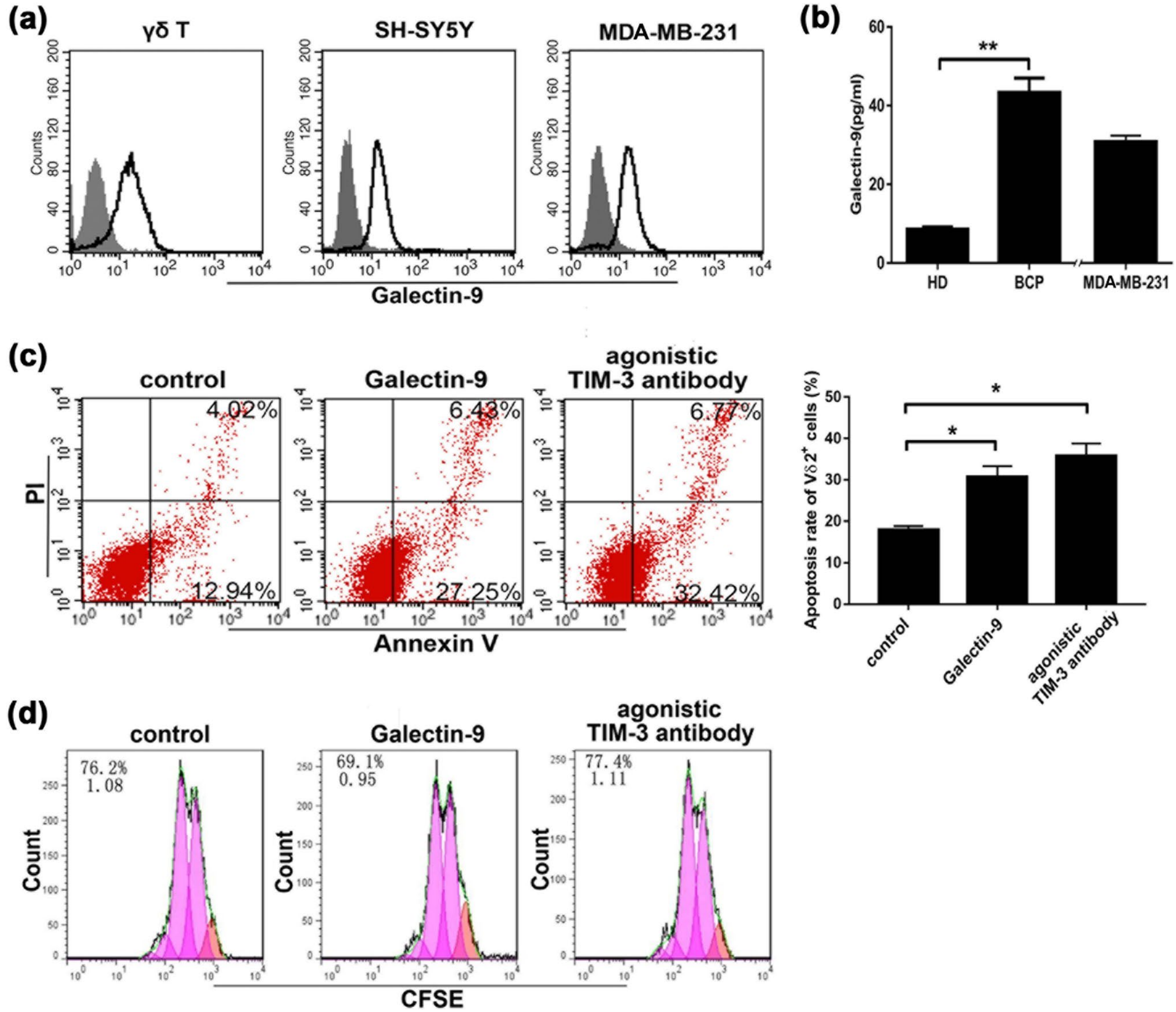

Fig. 2 Interaction of TIM-3 and TIM-3 ligands induced γδ T-cell dysfunction. **a** Intracellular detection of galectin-9 on γδ T cells, SH-SY5Y cells, and MDA-MB-231 cells. Shown is one representative data out of three. **b** The level of galectin-9 in the plasma of breast cancer patients (BCP, $n = 28$) and healthy donors (HD, $n = 20$) and the supernatant of MDA-MB-231 was determined by ELISA. Data are the mean ± SD ($n = 5$) of one representative experiment (**$p < 0.01$). **c** The isolated Vδ2+T cells from the γδ T-cell culture system with 24 h of galectin-9 or agonistic TIM-3 antibody treatment were stained with Annexin V-FITC and PI and the apoptosis rate of Vδ2+T cells were analyzed by flow cytometry. Shown is one representative experiment out of three. Columns represent the mean ± SD ($n = 3$) of one representative experiment out of three independent experiments (*$p < 0.05$). **d** After being treated by galectin-9 and agonistic TIM-3 antibody for 96 h, the proliferation ability of CFSE labeled Vδ2+T cells was determined by flow cytometry. The value (inset) for the percentage of cells that divided at least once (top left corner) and the average number of cell divisions (bottom left corner) are indicated for each sample. Similar results were obtained in three independent experiments

surface of these cells (data not shown). In addition, we found that galectin-9 also existed in human plasma and the level of this molecule in the plasma of breast cancer patients was significantly higher than that of healthy people (Fig. 2b). Besides, we detected a certain level of galectin-9 in the MDA-MB-231 culture supernatant (Fig. 2b). Therefore, we hypothesized that the apoptosis of the long-term cultured γδ T cell might be attributed to the interaction of TIM-3 and its ligands such as galectin-9. To address the function of the interaction of TIM-3 and galectin-9 on γδ T cells, we studied the effect of administration of galectin-9 or agonistic TIM-3 antibody on the survival and proliferation of γδ T cells expanded from breast cancer patients. We found that both galectin-9 and agonistic TIM-3 antibody treatment could significantly promote the apoptosis rate of γδ T cells when compared with control group without galectin-9 treatment

(Fig. 2c). However, the proliferation ability of γδ T cells was not influenced significantly by galectin-9 or agonistic TIM-3 antibody (Fig. 2d). These data indicated that the activation of TIM-3 pathway mainly influence the survival of γδ T cells but not their proliferation ability.

TIM-3 blockade reinvigorated dysfunctional γδ T cells

Many studies have reported that blocking one or both of the two major immune checkpoints CTLA-4 and PD-1 can efficiently reverse T-cell exhaustion and drugs targeting these checkpoints show promising clinical results in several cancer types [21, 22]. We supposed that γδ T-cell dysfunction could be reversed by TIM-3 blockade, since TIM-3 was found to be the major immune checkpoint related to γδ T-cell dysfunction. To address this question, we blocked TIM-3 signal activation using TIM-3 inhibitor or the fusion protein TIM-3-Fc which could neutralize TIM-3 ligands. After adding these two materials to the γδ T-cell culture system supplemented with galectin-9, respectively, and culturing for another 24 h, we determined the apoptotic status of γδ T cells. γδ T cells used in this experiment were expanded from one healthy donor and two breast cancer patients. One representative result is presented in Fig. 3a which showed that the proportion of annexin V positive γδ T cells in the groups treated with either TIM-3 inhibitor or TIM-3-Fc was significantly lower than the group without TIM-3 pathway blockade (Fig. 3a). To further confirm the effect of TIM-3 activation or blockade on the apoptosis and proliferation of γδ T cells, we have studied some signal transducers in terms of cell apoptosis or proliferation by western blotting and found that caspase 3 activation was involved in TIM-3 signaling activation which led to γδ T-cell apoptosis. However, the expression of cyclin B1 and cyclin D1 that correlated with cell proliferation was not influenced by either TIM-3 activation or its blockade (Fig. 3b). To further clarify the function of TIM-3 blockade, we compared the cytotoxicity of γδ T cells (expanded from breast

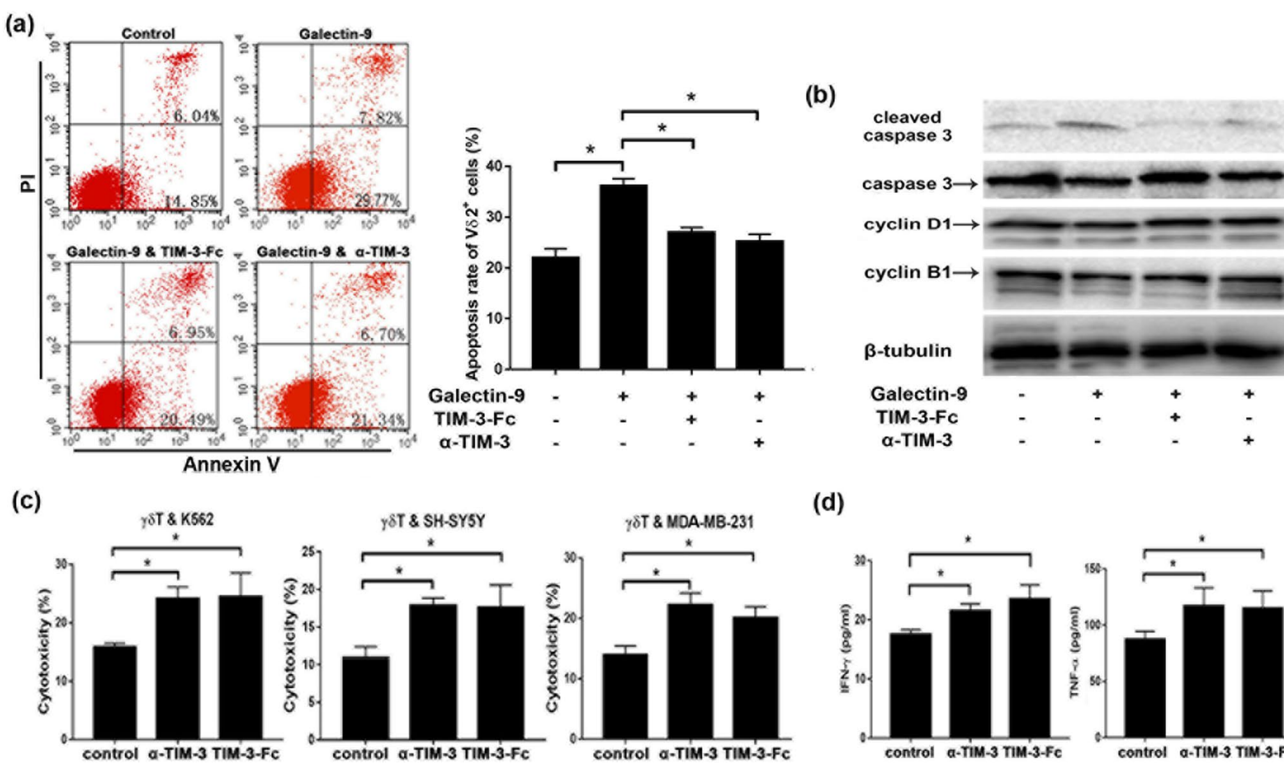

Fig. 3 TIM-3 blockade could reverse γδ T-cell dysfunction. **a** TIM-3 inhibitor (α-TIM-3), fusion protein TIM-3-Fc, or isotype control antibody was added to the γδ T-cell culture system (expanded from breast cancer patients) which have already supplied with galectin-9. After culturing for 24 h, the cells were stained with Annexin V-FITC and PI and analyzed by flow cytometry. Shown is one representative experiment out of three. **b** Western blot analyses. Galectin-9 activated caspase 3, while α-TIM-3 or TIM-3-Fc inhibits the activation of caspase 3 in γδ T cells. Galectin-9, α-TIM-3 or TIM-3-Fc could not influence the expression of cyclin B1 or cyclin D1 in γδ T cells. **c** γδ T cells (expanded from one breast cancer patient) with or without TIM-3 blockade (TIM-3 inhibitor or TIM-3-Fc) were co-cultured with different cell lines such as K562, SH-SY5Y and MDA-MB-231 at an *E*:*T* ratio of 1:1 for 12 h, then their cytotoxicity was determined by flow cytometry. **d** The levels of IFN-γ and TNF-α in each co-culture system mentioned above were determined by ELISA. Data are the mean ± SD ($n=3$) of one representative experiment out of three independent experiments (*$p<0.05$)

cancer patients for 18 day culture) with or without TIM-3 blockade by co-culturing γδ T cells with different cell lines such as K562, SH-SY5Y, and MDA-MB-231 for 12 h. As expected, either TIM-3 inhibitor or TIM-3-Fc could significantly enhance the cytotoxicity of γδ T cells compared with control group (Fig. 3c). Besides, we also determined the levels of IFN-γ and TNF-α in the γδ T cell and MDA-MB-231 co-culture system, and found that the co-culture system supplemented with either TIM-3 inhibitor or TIM-3-Fc had significantly higher levels of IFN-γ and TNF-α than control group (Fig. 3d). Collectively, TIM-3 pathway blockade could efficiently inhibit the apoptosis of γδ T cells and enhance their cytotoxicity but not influence the proliferation of γδ T cells.

MT110 enhanced γδ T cells mediated cytotoxicity and up-regulated TIM-3 in vitro

γδ T cells kill cancer cells mainly by two mechanisms: one is direct lysis by cell to cell contact and the other is through the secreted cytokines such as IFN-γ and TNF-α [23, 24]. Recent studies found that BiTE® molecules could strengthen T-cell cytotoxicity through these two manners both in vitro and in vivo [17, 25, 26]. As one of the most frequently and intensely expressed tumor-associated antigens known, EpCAM is expressed on a great variety of human adenocarcinoma and squamous cell carcinoma [27]. Our results showed that EpCAM was highly expressed on the surface of the breast cancer cell MDA-MB-231, while hardly expressed on neuroblastoma cell SH-SY5Y (Fig. 4a). Hence, EpCAM

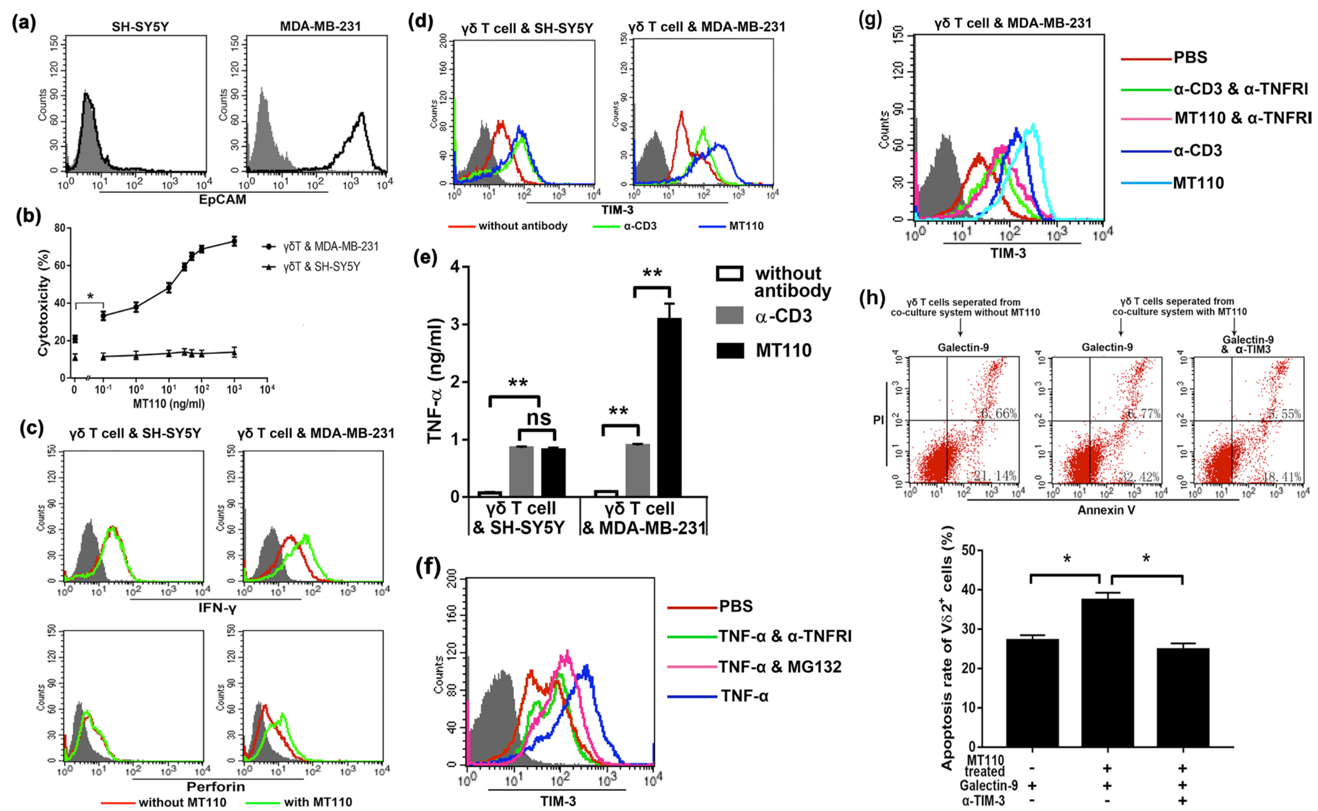

Fig. 4 MT110 enhanced γδ T-cell-mediated cytotoxicity and up-regulated TIM-3 in vitro. **a** Surface detection of EpCAM on the neuroblastoma cell line SH-SY5Y and the breast cancer cell line MDA-MB-231. **b** A series dose of MT110 was added to the γδ T cell and SH-SY5Y or MDA-MB-231 co-culture system ($E{:}T=5{:}1$) and the cytotoxicity of γδ T cell was determined by flow cytometry after 12 h culture. **c** Intracellular detection of IFN-γ and perforin on γδ T cells co-cultured with SH-SY5Y or MDA-MB-231 at an $E{:}T$ ratio of 5:1 supplying MT110 (30 ng/ml) or not. **d** The expression level of TIM-3 on γδ T cells in the co-culture system with anti-CD3 antibody or MT110 treatment or without any treatment was determined by flow cytometry after 24 h co-culturing. Shown is one representative experiment out of three. **e** The levels of IFN-γ and TNF-α in each co-culture system mentioned above were determined by ELISA. **f** γδ T cells were treated with or without TNF-α inhibitor (α-TNFRI) or NF-κB inhibitor (MG132) in the presence of 3 ng/mL TNF-α for 24 h. **g** The γδ T and MDA-MB-231 co-culture system was treated with or without TNF-α inhibitor or NF-κB inhibitor for 24 h. The expression of TIM-3 was measured by flow cytometry. Shown is one representative experiment out of three. **h** TIM-3^{high}γδ T cells separated in the co-culture system were treated by galectin-9 (40 pg/ml) in the presence or absence of TIM-3 inhibitor (5 μg/ml) for 24 h, and then, the cells were stained with Annexin V-FITC and PI and the apoptosis was analyzed by flow cytometry. Shown is one representative experiment out of three. All data mentioned above are the mean ± SD ($n=3$) of one representative experiment out of three independent experiments (*$p<0.05$, **$p<0.01$)

was selected as the immunotarget and the BiTE®, MT110, which can bind to CD3 and EpCAM simultaneously was utilized in the following experiments accordingly. We examined the influence of MT110 on γδ T cells (expanded from breast cancer patients) mediated cytotoxicity by co-culturing γδ T cells with MDA-MB-231 or SH-SY5Y cells at an effector: target ratio of 5:1. A representative result showed that MT110 was able to strengthen the cytolytic effect of γδT cells against MDA-MB-231 cells in a dose dependent manner and the cytotoxicity exceeded 60% when the concentration of MT110 exceeded 30 ng/ml. However, the cytolytic effect of γδT cells against SH-SY5Y cells was not significantly enhanced even the concentration of MT110 reached to 1000 ng/ml (Fig. 4b). We also analyzed cytokine production ability of γδ T cells in response to MT110 in the presence of MDA-MB-231 or SH-SY5Y. As expected, γδ T cells produced higher levels of IFN-γ and perforin in the presence of MT110 than those in the absence of MT110. However, for the γδ T cell and SH-SY5Y co-culture system, the cytokines production ability of γδ T cells in response to MT110 was not strengthened (Fig. 4c). These results suggested that MT110 could further activate γδ T cells and enhance their cytotoxicity in the presence of EpCAM-positive target cells.

Since γδ T cells were further activated by MT110 in the presence of MDA-MB-231, we examined the influence of MT110 treatment on TIM-3 expression on γδ T cells in the co-culture system. Our results showed that both α-CD3 and MT110 treatment could up-regulate the expression of TIM-3 on γδ T cells in the presence of SH-SY5Y cells, although MT110 treatment did not further enhance the expression of TIM-3 compared with α-CD3 treatment. However, in the presence of MDA-MB-231 cells, MT110 could further elevate the expression level of TIM-3 on γδ T cells compared with α-CD3 which was widely used to activate T cells (Fig. 4d). Thus, another molecular mechanism to induce TIM-3 expression in γδ T cells than stimulation via CD3 activation may exist. Zheng et al. reported that that TNF-α could induce TIM-3 expression on NK cells [28]. Then, we examined the TNF-α in the co-culture system with or without α-CD3 or MT110 treatment and explored its function on the expression of TIM-3 on γδ T cells. Results showed that although both α-CD3 and MT110 treatment could significantly elevate the level of TNF-α in the co-culture system, MT110 treatment could further elevate the level of TNF-α in the γδ T and MDA-MB-231 co-culture system compared with α-CD3 (Fig. 4e). These results are consistent with the level of TIM-3 in Fig. 4d. To address the role of TNF-α on TIM-3 expression on γδ T cells, we treated γδ T cells by recombinant human TNF-α protein for 24 h and found that TNF-α could enhance the expression of TIM-3 on γδ T cells (Fig. 4f). To further confirm the effect of TNF-α on the expression of TIM-3, we added TNF-α inhibitor (anti-TNFRI blocking antibody) or NF-κB (a well-known transcription factor that is activated by TNF-α) inhibitor to γδ T-cell culture. After 24 h treatment, either TNF-α inhibitor or NF-κB inhibitor could attenuate TNF-α-induced TIM-3 expression (Fig. 4f). In addition, we examined the effect of TNF-α inhibitor on the α-CD3 or MT110-induced TIM-3 expression in the presence of MDA-MB-231 cells. As expected, TNF-α inhibitor could mitigate the up-regulation of TIM-3 on γδ T cells induced by α-CD3 or MT110 (Fig. 4g). These results indicated that except for CD3 activation, MT110 could further up-regulate TIM-3 expression on γδ T cells through the further increased level of TNF-α in the co-culture system with MDA-MB-231 cells.

Furthermore, we found that these TIM-3^{high} γδ T cells separated in the co-culture system with MDA-MB-231 cells were more sensitive to apoptosis induced by galectin-9 and the apoptosis of these TIM-3^{high} γδ T cells could be significantly inhibited by TIM-3 inhibitor (Fig. 4h). These results suggested that with the enhanced activation and strengthened cytotoxicity of γδ T cells by MT110 treatment, the expression of TIM-3 was also further up-regulated. Although the TIM-3 up-regulated γδ T cells were apt to apoptosis, this apoptosis could be also efficiently mitigated by TIM-3 blockade.

Combination of TIM-3 inhibitor and MT110 enhanced the survival and infiltration ability of transfused γδ T cells in vivo

To determine the influence of TIM-3 inhibitor and MT110 on the activity of adoptively transfused γδ T cells in vivo, we transfused the ex vivo amplified γδ T cells (expanded from a breast cancer patient) with or without TIM-3 inhibitor and/or MT110 to breast cancer xenograft mice model and explored the distribution and accumulation status of these cells. Each breast cancer mouse in γδ T group, γδ T and MT110 group and γδ T and α-TIM-3 and MT110 group received 5×10^6 PKH26 labeled γδ T cells with or without α-TIM-3 and/or MT110, respectively. At the time point of 24 or 120 h after transfusion, we detected the proportion of the transfused γδ T cells in the peripheral blood of each breast cancer mouse by flow cytometry. Results showed that the proportion of PKH26-positive cells in the peripheral blood of cancer mice in both γδ T and MT110 group and γδ T and α-TIM-3 and MT110 group was significantly lower compared with γδ T group after 24 h transfusion. Whereas, there were no apparent differences between γδ T and MT110 group and γδ T and α-TIM-3 and MT110 group (Fig. 5a, b). Besides, at the time point of 120 h after transfusion, the differences in the proportion of PKH26-positive cells in peripheral blood among these three groups were not significant (Fig. 5a, b).

Simultaneously, for the same three groups of breast cancer mice, the infiltration and accumulation status of the transfused γδ T cells at tumor sites was studied by measuring

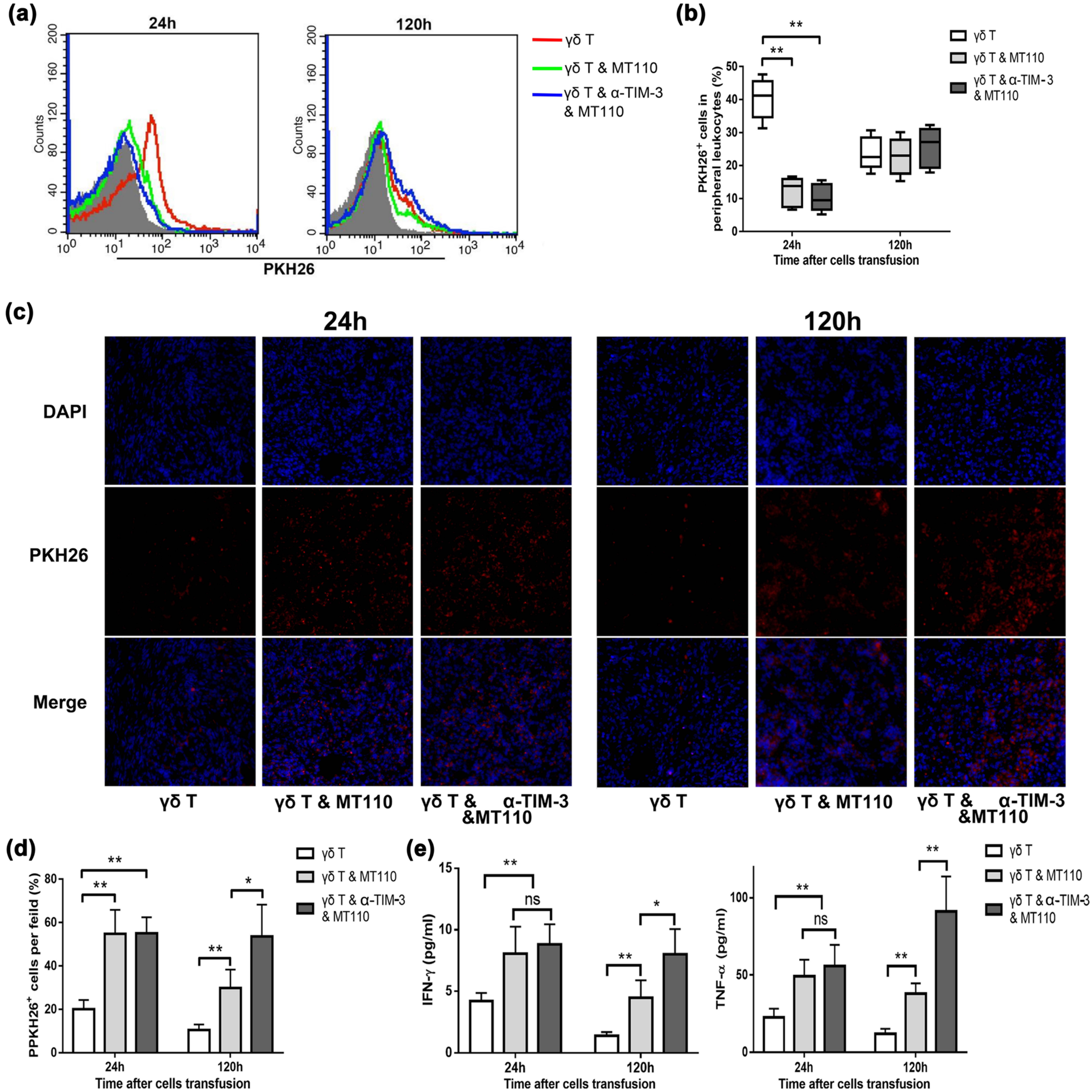

Fig. 5 Combination of TIM-3 inhibitor and MT110 enhanced the survival and infiltration ability of adoptively transfused γδ T cells in vivo. **a** 24 or 120 h after breast cancer mice in γδ T group ($n=6$), γδ T and MT110 group ($n=6$) and γδ T and α-TIM-3 and MT110 group ($n=6$) receiving transfusion of PKH26 labeled γδ T cell (expanded from a breast cancer patient) with or without α-TIM-3 and MT110, respectively, the proportions of PKH26 positive cells in the peripheral leukocytes of breast cancer mice in each group were determined by flow cytometry (gray: blank; Red: γδ T group; green: γδ T and MT110 group; blue: γδ T and α-TIM-3 and MT110 group). Shown is one representative experiment. **b** Comparison of proportions of PKH-positive cells in the peripheral leukocytes of breast cancer mice between the three groups mentioned above at the time point of 24 or 120 h after transfusion. (**$p<0.01$). Whiskers in box plots indicate maximum and minimum values measured. Line indicates the median. **c** Accumulation of transfused γδ T cells (purple) in tumor tissues of three different groups mentioned above as visualized by DAPI blue and PKH26 red double labeling at the time point of 24 or 120 h after γδ T-cell transfusion (×200). Shown is one representative data out of six. **d** Number of DAPI blue and PKH26 red double-labeled transfused γδ T-cell count for each microscopic viewing field in shown. Bars indicate mean count for each group mentioned above and error bars are indicated. (*$p<0.05$; **$p<0.01$). **e** The plasma of breast cancer mice in each group was collected and IFN-γ and TNF-α levels were determined by ELISA. Data are the mean ± SD ($n=5$) of one representative experiment (*$p<0.05$; **$p<0.01$)

the density of PKH26-positive cells in pathological frozen section. Results showed that the density of PKH26-positive cells in the tumor tissue of both γδ T and MT110 group and γδ T and α-TIM-3 and MT110 group was significantly higher than that of γδ T group at the time point of 24 h after transfusion. Whereas, there were no significant differences between the γδ T and MT110 group and γδ T and α-TIM-3 and MT110 group (Fig. 5c, d). However, at the time point of 120 h after transfusion, the density of PKH26-positive cells in the tumor site of the γδ T and α-TIM-3 and MT110 group was significantly higher compared to the γδ T and MT110 group (Fig. 5c, d). This might be due to the apoptosis inhibitory function of TIM-3 inhibitor.

In addition, we determined the levels of cytokines such as IFN-γ and TNF-α in the plasma of the tumor mice at the time point of 24 or 120 h after γδ T-cell transfusion. As expected, the levels of IFN-γ and TNF-α in either γδ T and MT110 group or γδ T and α-TIM-3 and MT110 group were significantly higher than those of γδ T group at both of the time points. Although there were no significant differences in the levels of these two cytokines between γδ T and MT110 group and γδ T and α-TIM-3 and MT110 group at 24 h after transfusion, γδ T and α-TIM-3 and MT110 group exhibited significantly higher levels of both IFN-γ and TNF-α compared with γδ T and α-TIM-3 group at 120 h after transfusion (Fig. 5e). These data were consistent with the results mentioned above.

Collectively, these results indicated that MT110 could promote the adoptively transfused γδ T cells to infiltrate into tumor tissues from peripheral blood and accumulate in tumor tissues. Furthermore, with the help of TIM-3 inhibitor, the accumulated γδ T cells in tumor tissues could resist apoptosis induced by TIM-3 ligands in tumor environment and survived for a longer period.

Combined application of TIM-3 inhibitor and MT110 further enhanced anti-tumor effect of adoptively transfused γδ T cells in vivo

Building on previous data demonstrating γδ T cells as effective candidates for tumor immunotherapy [29, 30] and the results mentioned above, we hypothesized that the tumor therapeutic efficacy of γδ T cells could be further enhanced by combining them with TIM-3 inhibitor and the bispecific antibody MT110. To test this hypothesis, MDA-MB-231 inoculated BALB/C nude mice were received three cycles of γδ T cells (expanded from a breast cancer patient) transfusion in combination with α-TIM-3 and/or MT110 in corresponding groups. At the beginning of the first cycle of adoptively transfusion, we began to measure tumor volume of MDA-MB-231-bearing mice in each group every 2 days. According to the tumor index analysis, the tumor volume of MDA-MB-231-bearing mice in the groups received γδ T-cell transfusion alone or with α-TIM-3 and/or MT110 was significantly smaller compared with PBS group. Furthermore, the tumor volume of the γδ T and α-TIM-3 and MT110 group was even significantly smaller compared with all the other groups (Fig. 6a). These data indicated that γδ T-cell transfusion in combination with α-TIM-3 and MT110 could significantly further enhance the tumor therapeutic efficacy of γδ T cells. Besides, the median survival periods of the PBS group, γδ T group, γδ T and α-TIM-3 group, and γδ T and MT110 group were 35, 45, 50, and 52 days, respectively, while the mice in the γδ T and α-TIM-3 and MT110 group had a median survival period of 60 days which was significantly longer than all other groups (Fig. 6b). These results demonstrated that combined application of TIM-3 inhibitor and MT110 could further strengthen the anti-tumor efficacy of the adoptively transfused γδ T cells.

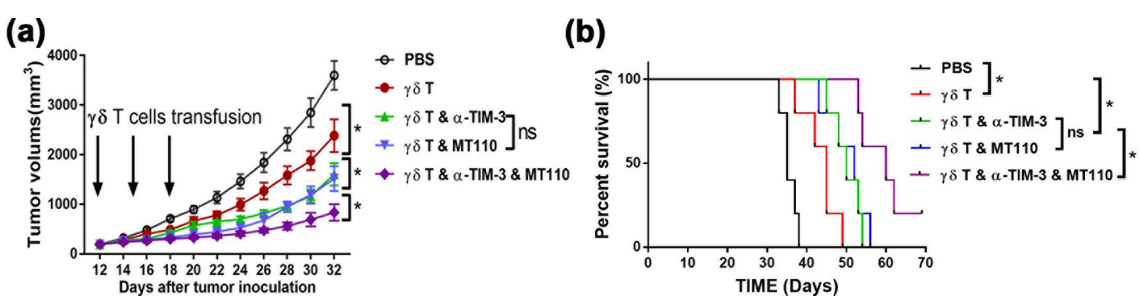

Fig. 6 Combined application of TIM-3 inhibitor and MT110 further enhanced anti-tumor effect of the adoptively transfused γδ T cells in vivo. **a** PBS or γδ T cells (expanded from a breast cancer patient) with or without α-TIM-3 or/and MT110 was intravenously injected into the tail of mice in the appropriate group, respectively, at day 12, day 15, and day 18. Shown is tumor growth curve of breast cancer mice in each group. Tumor sizes are plotted as mean ± SD for each group ($n = 5$). Experiments were repeated two to three times with similar results. (*$P < 0.05$). **b** The median survival periods of mice in γδ T and α-TIM-3 and MT110 group were 60 days, while PBS group, γδ T group, γδ T and α-TIM-3, and γδ T and MT110 group were 35, 45, 50, and 52 days, respectively. The differences were statistically significant. (*$P < 0.05$)

Discussion

The present study has addressed two main issues regarding the activity of ex vivo expanded γδ T cells for tumor immunotherapy: first, to show blockade of the key immune checkpoint could inhibit γδ T-cell dysfunction and reinvigorate their effector function; second, to show that a bispecific antibody, especially a BiTE® could promote γδ T cells to localize to tumor sites and further enhance γδ T-cell cytotoxicity.

Since Ensslin and collogues found that γδ T-cell cytotoxicity against a range of tumor cell lines is greater than that achievable with αβ T cells which have been the focus of most trials of tumor adoptive immunotherapy, γδ T cells have attracted more and more scientists' attention for tumor immunotherapy [31]. However, as a kind of immunotherapy, γδ T-cell-based immunotherapy also confronted several unfavorable restraints, one of which is γδ T-cell dysfunction. In this study, we found that with the prolongation of induction and expansion in vitro, γδ T cells began to become exhausted gradually when they reached the summit of cytotoxicity at about day 15. These exhausted γδ T cells exhibited a weakened cytotoxicity and increased sensitivity to apoptosis. This phenomenon also exists in the ex vivo expansion of other immune cells such as CIK cells and NK cells. This is also the reason why most basic studies or clinical trials began to harvest CIK cells or NK cells around 14 day cultivation and then performed adoptive transfusion [32, 33]. However, we found that the levels of activation markers such as CD69 and HLA-DR on γδ T were still increasing after their cytotoxicity reached to summit at day 15. This might be due to that γδ T cells had already begun to apoptosis gradually from day 14, although their activation markers were still increasing in the following or that the long-term hyper-activation induced γδ T-cell apoptosis. In addition, the transfused γδ T cells will become exhausted in tumor microenvironment in vivo and, thereafter, their immunotherapeutic effect will be inevitably weakened. Therefore, preventing or reversing immune cell dysfunction has become an important measure to improve the tumor immunotherapeutic efficacy of the adoptively transfused immune cells recently.

Under normal physiological conditions, immune checkpoints are crucial for the maintenance of self-tolerance and also to protect normal tissues from damage when the immune reaction is excessive. However, the expression of immune checkpoints can be dysregulated by tumors and help tumor cells escape from immune surveillance. It has been noted that dysregulated and sustained expression of immune checkpoint molecules marks the impaired or exhausted T cells that exist in chronic disease states such as cancer [13, 34]. To identify the factors that contribute to γδ T-cell dysfunction, the dynamic expression of some immune checkpoints such as CTLA-4, PD-1, and TIM-3 on γδ T cells during their induction and expansion process was examined in this study. We found that TIM-3 had a rapid and robust up-regulation on γδ T cells, while the changes of the expression of CTLA-4 and PD-1 were not obvious, although CTLA-4 had a rapid up-regulation in the early days. However, the results of the dynamic expression of PD-1 in our study were not consistent with the report from Iwasaki et al., since they found that PD-1 was up-regulated significantly during early stage of γδ T-cell expansion in vitro [35]. We supposed that different activators and concentrations used in the γδ T-cell culture system might contribute to the different results, since 10 μM HMB-PP was used to induce γδ T cells from PBMC in Iwasaki's study, while 1 μM zoledronate was used in our study. In addition, the following experiments in our study suggested that TIM-3 might be an important immune checkpoint related to γδ T-cell dysfunction. Indeed, several studies also found that TIM-3 was up-regulated on exhausted αβ T cells in both chronic infection diseases and human cancer [34, 36].

As one of the ligands of TIM-3, galectin-9, was broadly distributed in various tissues and exclusively highly expressed in the developing thymus, where it conceivably plays an essential role in the clonal deletion of T cells via thymic epithelial cell–thymocyte interactions leading to apoptosis [19]. Regarding its expression in cancer, galectin-9 was identified as a tumor antigen in Hodgkin's lymphoma; however, several groups have shown that galectin-9 levels vary when comparing tumor tissue and normal tissue [37, 38]. This might be due to this molecule which has several splice variants and the galectin-9 level detected among different studies might be different splice variants [39]. Besides, galectin-9 are localized on the cell membrane, in the cytoplasm and the nucleus of cells [40]. This further leads to inconsistency between different studies. In this study, we found that galectin-9 was existed in the cytoplasm of γδ T cells and MDA-MB-231 cells but not expressed on their membrane. Besides, we also found that a certain amount of galectin-9 existed in the supernatant of MDA-MB-231 cells. Moreover, in the plasma of breast cancer patients, we detected a significant high level of galectin-9 when comparing with health women. Many studies have reported that the interaction of galectin-9-TIM-3 could not only induce apoptosis in a number of immune cell types like Th1 cells and CD8 + cytotoxic T (Tc) cells, but also play an important role in the immune suppression mediated by T regulatory cells (Tregs) [15, 41, 42]. All of these consequences contribute to tumor immune escape and promote tumor progression. Therefore, blockade of galectin-9-TIM-3 pathway might be a very reasonable treatment to improve tumor immunity.

In this study, we found that the activation of TIM-3 could markedly promote γδ T-cell apoptosis, while interrupting galectin-9-TIM-3 pathway could significantly mitigate the apoptosis of γδ T cells. It is noteworthy that TIM-3 pathway mainly influenced early apoptosis of γδ T cells but not late apoptosis, since the proportion of annexin V and PI double positive γδ T cells was almost unchanged by TIM-3 activation or blockade. For the study of signal transductions in terms of apoptosis induced by TIM-3 signaling, we studied some pathways that related to apoptosis and found that caspase 3 activation was involved in TIM-3 signaling pathway which led to γδ T-cell apoptosis, although the specific complete signaling pathway was not identified in this study.

Dysfunctional T cells often display a hypo-proliferative ability; however, TIM-3 pathway blockade by either TIM-3 inhibitor or TIM-3-Fc fusion protein did not show any significant enhancement of the proliferation ability of γδ T cells in this study. This result was not in accordance with Zhu and colleagues' study which displayed that TIM-3-Fc fusion protein could induce Th1 cells hyperproliferation [15]. This might be due to that TIM-3 signal pathway does not participate in the regulation of the proliferation of γδ T cells. Since impaired cytotoxicity is another common feature of exhausted immune cells, the influence of TIM-3 pathway blockade on the cytotoxicity of γδ T cells was also determined in this study and the results showed that both TIM-3 inhibitor and TIM-3-Fc fusion protein could significantly enhanced γδ T-cell cytotoxicity. Since TIM-3 pathway blockade had almost no influence on the proliferation of γδ T cells, the enhanced cytotoxicity might be due to the mitigated apoptosis and the promotion of cytokines production. To block TIM-3 pathway efficiently, TIM-3 inhibitor was selected as one of the components for the tumor immunotherapy strategy in the following animal experiments.

Since CD3 is also expressed by γδ T cells and participates in the activation signal transduction of γδ T cells, we utilized the BiTE® molecule, MT110, to maximize γδ T cell-mediated cytotoxicity and promote γδ T cells to accumulate in tumor tissue in this study. This small, tandem scFv format could significantly reduce the intercellular space between T cells and cancer cell membranes, and activate $CD4^+$ T or $CD8^+$ T cells in the presence of target cells, enabling the formation of immunological cytolytic synapses and, therefore, mediating lysis of the attached target cells [26, 43]. In this study, we found that MT110 could significantly enhance γδ T-cell cytotoxicity even at 0.1 ng/ml and this enhancement became strengthened with the increasing concentration. When the concentration reached to 30 ng/ml, the cytotoxicity even exceeded 60%. When the concentration increased from 30 to 1000 ng/ml, the cytotoxicity increased only to about 80%. This suggested that when the treatment concentration of MT110 reached to a certain level, the cytotoxicity could not be further enhanced. This might be due to the binding capacity of this bispecific antibody in the co-culture system which has reached saturation. Therefore, we selected 30 ng/ml in the following ex vivo experiment, and found that MT110 treatment could enhance the production of IFN-γ and perforin of γδ T cells in the presence of MDA-MB-231. However, MT110 could neither enhance the cytotoxicity nor the cytokine production of γδ T cells when co-cultured with SH-SY5Y. This might be due to MDA-MB-231 cells had a high level of EpCAM, while SH-SY5Y cells were EpCAM negative. Therefore, γδ T cells could be further activated by target positive cells which served as feeder cells especially with the help of bispecific antibody. However, with the further activation by MT110, the immune checkpoint TIM-3 was also significantly up-regulated on γδ T cells. Furthermore, the TIM-3^{high} γδ T cells treated by MT110 were more vulnerable to apoptosis induced by galectin-9 and the apoptosis could be significantly alleviated by TIM-3 inhibitor. These results further confirmed the rationality of the combined application of TIM-3 inhibitor and MT110.

Although many kinds of immune cells such as CIK and NK cells have shown very strong ability to kill tumor cells in vitro, their clinical therapeutic efficacy was not so good as intended. Because there are many factors limit the tumor therapeutic efficacy of the adoptively transfused immune cells. First, tumor cells can express or release a large number of molecules or cytokines that can bind to the co-inhibitory receptors expressed on T cells, leading to apoptosis of T cells. Second, many immune suppressive cells such as myeloid derived suppressor cells (MDSC), Tregs, and tumor-associated macrophages (TAMs) exist in tumor microenvironment and can suppress the function of immune cells [44–46]. Third, the adoptively transferred immune cells were difficult to migrate toward tumor sites and infiltrate into them. In this study, with the help of MT110 and TIM-3 inhibitor, the transfused γδ T cells could efficiently infiltrate into tumor tissues and then accumulate in them. Furthermore, with the blocking of TIM-3, the transfused γδ T cells survived for a significantly longer period in tumor sites. In the following cancer therapy experiment in vivo, there was no significant difference between γδ T and α-TIM-3 group and γδ T and MT110 group, though either anti-TIM-3 blocking antibody or MT110 could significantly enhanced the anti-tumor effect of the transfused γδ T cells. However, when TIM-3 inhibitor and MT110 were combined together with γδ T-cell transfusion, the anti-tumor function of γδ T cells could be significantly further enhanced. Obviously, these in vivo tumor treatment outcomes achieved our desired objective and were consistent with the previous experiment results.

In conclusion, our study demonstrated that the immune checkpoint TIM-3 was the key factor that related to γδ T-cell dysfunction. Blocking the binding of TIM-3 to its ligands such as galectin-9 by TIM-3 inhibitor or TIM-3-Fc could effectively reinvigorate the dysfunctional γδ T-cell in vitro. With the help of the bispecific antibody MT110, γδ T cells exhibit strengthened cytotoxicity to breast cancer cells in vitro and enhanced accumulation in tumor sites in vivo. When combined TIM-3 inhibitor and MT110 with γδ T-cell adoptive transfusion, the anti-tumor activity of the transfused γδ T cells was further strengthened. With the better understanding of the factors relative to γδ T-cell dysfunction and accumulation, we can improve the quality of ex vivo expanded γδ T cells and make full use of them to treat cancer patients.

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Author contributions QG, PZ, and Zhen Z performed experiments. QG, JZ, Zheng Z, and LH analyzed data. YH, BH, NL, and XZ assisted with experiments. QG and LH conceived and designed the project. QG and LH wrote the manuscript.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no potential conflict of interest.

Ethical approval This study was approved by the ethical review board of Qingdao University, and the care and use of animals was conducted according to protocols approved by the Qingdao University Institutional Animal Care and Use Committee. All procedures performed in studies involving human participants were in accordance with the research ethics committee of Qingdao University and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

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